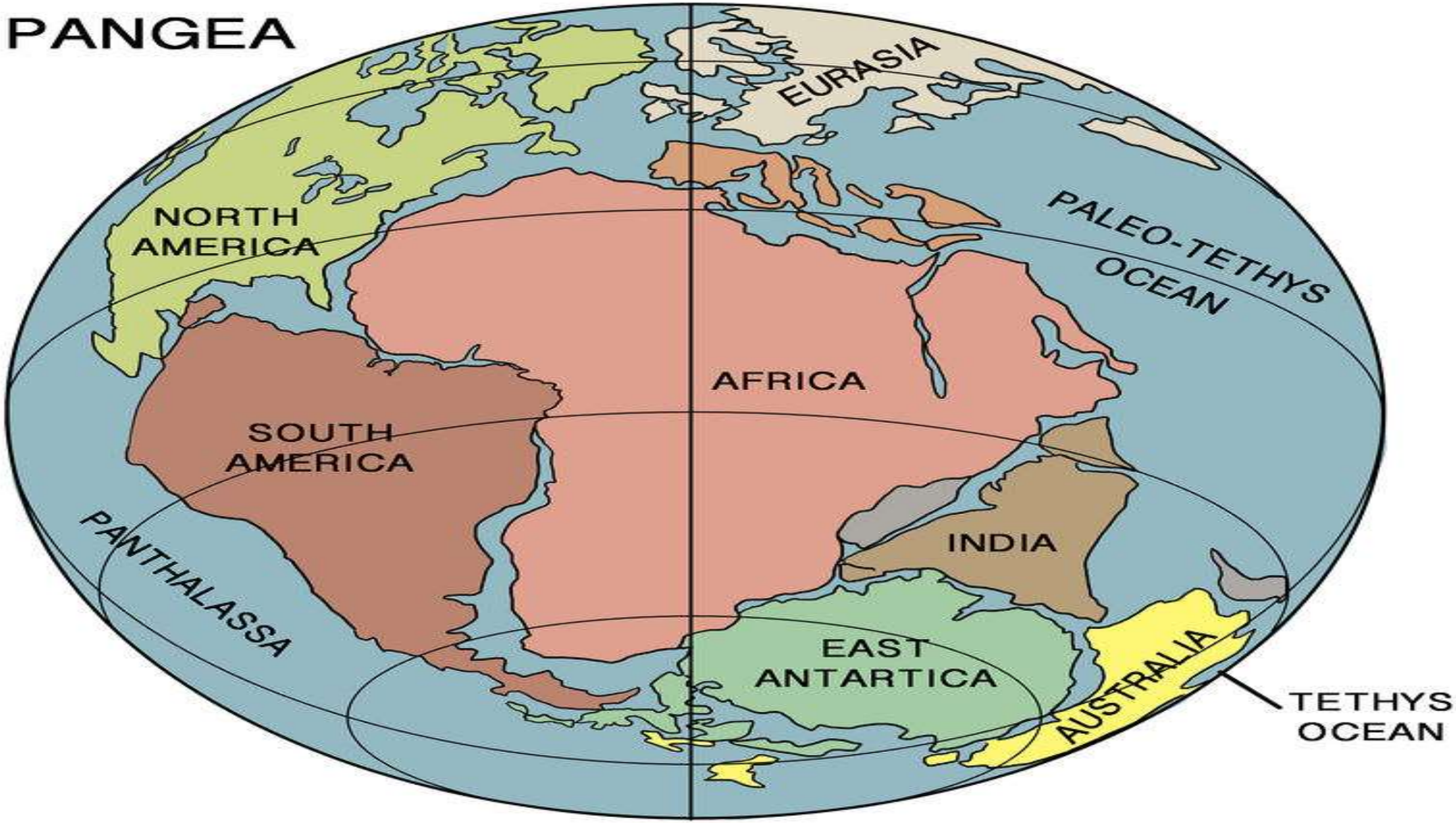
# Continental Drifting


PANGEA
EURASIA
NORTH AMERICA
PALEO-TETHYS OCEAN
AFRICA
SOUTH AMERICA
INDIA
PANTHALASSA
EAST ANTARTICA
AUSTRALIA
TETHYS OCEAN

# Building the Continents

- The Earth formed 4.5 billion years ago from a great, swirling cloud of dust and gas. The continuous smashing of space debris and the pull of gravity made the inside of Earth heat up. As the heat increased, some of Earth's rocky materials melted and rose to the surface, where they cooled and formed a crust. Heavier material sank toward Earth's center. Eventually, the earth came to have three main layers: the core, the mantle, and the crust

- The crust and the top portion of the mantle form a rigid shell around the earth that is broken up into huge sections called tectonic plates. The heat from inside the earth causes the plates to slide around on the molten mantle. Today, tectonic plates continue to slowly slide around the surface, just as they have been doing for hundreds of millions of years. Geologists believe the interaction of the plates, a process called plate tectonics, contributed to the creation of continents.

# CONTIFNENTAL DRIFT

- The face of the Earth, that is, its visible surface has undergone radical changes in the past. Geologists explained these changes as the consequences of the cooling and contraction of the earth, through thousands of years. This explanation seemed quite unsatisfactory to a German scientist, ALFRED WEGENER (1880-1930)

- In 1915, Wegener published a book ‘THE ORIGIN OF CONTINENTS AND OCEANS’ in which he advanced a new theory, the theory of CONTINENTAL DRIFT.

- This theory claimed that the changes in the appearance of the earth were, mainly due to the shifting of continents. Wegener grounded his theory primarily on two premises.

- (i) First, that the geological formation and fossil remains of the present far away continents showed striking similarities.

- (ii) Second, that some of the continents showed astonishingly complementary coastlines. The east coast of South America, for example, matches the west coast of Africa, so finally that they would fit together exactly, if they were brought together.

# Earth’s seven main divisions of land.

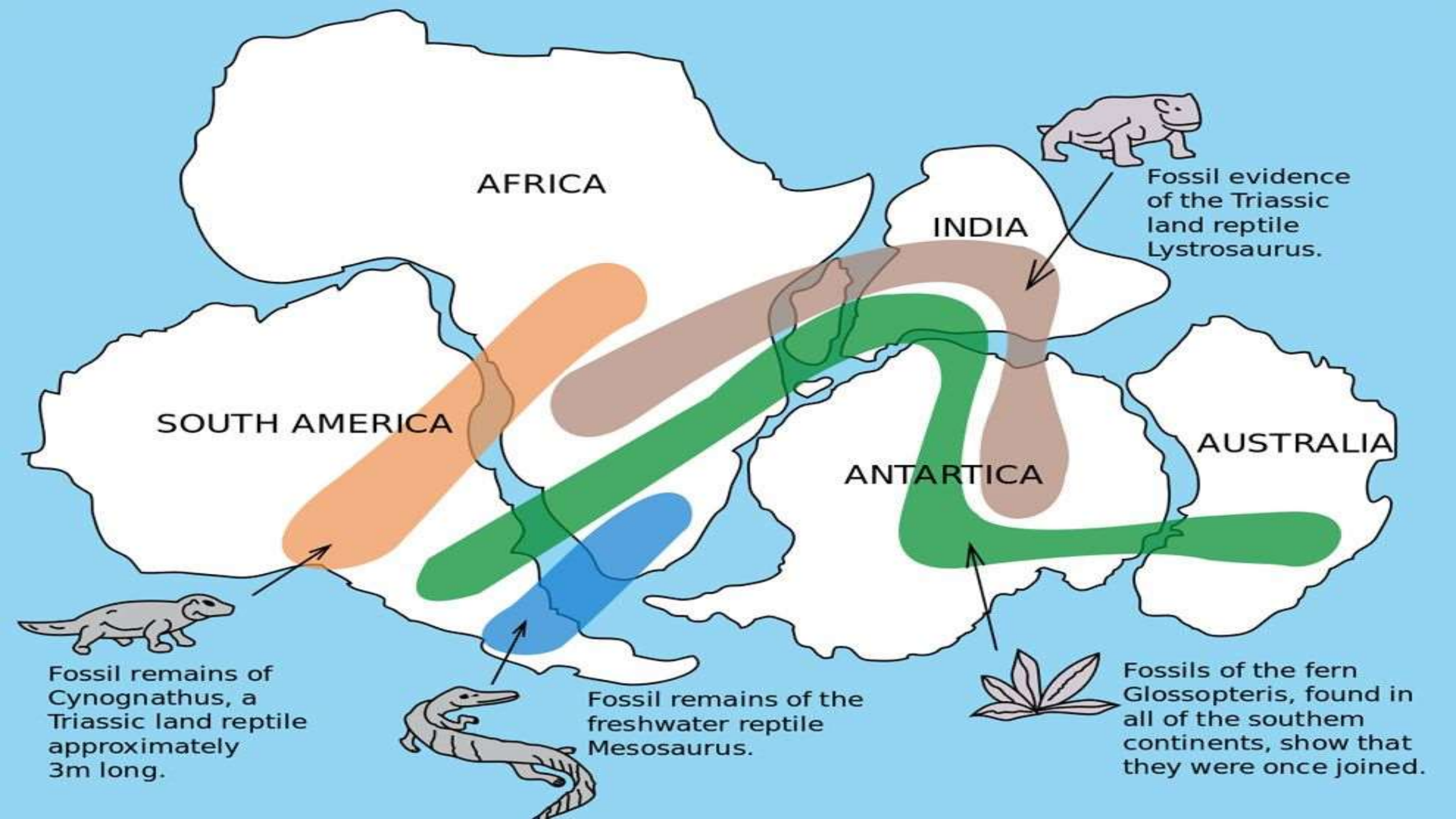
- The continents are, from largest to smallest: Asia, Africa, North America, South America, Antarctica, Europe, and Australia.

- When geographers identify a continent, they usually include all the islands associated with it. Japan, for instance, is part of the continent of Asia. Greenland and all the islands in the Caribbean Sea are usually considered part of North America.

- Together, the continents add up to about 148 million square kilometers  of land. Continents make up most—but not all—of Earth’s land surface. A very small portion of the total land area is made up of islands that are not considered physical parts of continents. The ocean covers almost three-fourths of Earth. The area of the ocean is more than double the area of all the continents combined. All continents border at least one ocean. Asia, the largest continent, has the longest series of coastlines.


PANTHALASSIC
OCEAN
Wrangellia
NORTH
AMERICA
PANGEA
EURASIA
China
Cimmeria
TETHYS OCEAN
SOUTH
AMERICA
GONDWANA
AFRICA
India
AUSTRALIA
ANTARCTICA


WORLD MAP
Greenland (Denmark)
Alaska (USA)
Canada
United States of America
Mexico
Russia
China
India
Australia
Kazakhstan
Mongolia
Brazil
Peru
Bolivia
Paraguay
Chile
Argentina
Uruguay
Colombia
Venezuela
Ecuador
Guyana
Suriname
French Guiana (Fr.)
Indonesia
Malaysia
Singapore
Brunei
Timor-Leste
Papua New Guinea
Philippines
Vietnam
Thailand
Myanmar (Burma)
Bangladesh
Bhutan
Nepal
Sri Lanka
Maldives
Pakistan
Afghanistan
Iran
Iraq
Saudi Arabia
Yemen
Oman
United Arab Emirates
Qatar
Bahrain
Kuwait
Jordan
Israel
Turkey
Georgia
Armenia
Azerbaijan
Turkmenistan
Uzbekistan
Kyrgyzstan
Tajikistan
North Korea
South Korea
Japan
Taiwan
Laos
Sweden
Finland
Norway
Iceland
Faroe Islands (Denmark)
Ireland
United Kingdom
Denmark
Estonia
Latvia
Lithuania
Belarus
Poland
Germany
Ukraine
France
Spain
Portugal
Andorra
Monaco
San Marino
Vatican City
Italy
Switzerland
Austria
Hungary
Romania
Bulgaria
Greece
Malta
Tunisia
Morocco
Western Sahara
Algeria
Libya
Egypt
Mauritania
Mali
Niger
Chad
Sudan
Eritrea
Djibouti
Ethiopia
Somalia
Senegal
The Gambia
Cabo Verde
Guinea-Bissau
Guinea
Sierra Leone
Liberia
Côte d'Ivoire
Burkina Faso
Nigeria
Central African Rep.
South Sudan
Uganda
Kenya
Rwanda
Burundi
Tanzania
Dem. Rep. of the Congo
Gabon
Congo
São Tomé and Príncipe
Equatorial Guinea
Cameroon
Angola
Zambia
Zimbabwe
Namibia
Botswana
Mozambique
Madagascar
Comoros
Mayotte (Fr.)
Seychelles
Mauritius
Reunion (Fr.)
eSwatini
Lesotho
South Africa
Haiti
Dominican Rep.
Puerto Rico (U.S.)
The Bahamas
Cuba
Jamaica
Belize
Guatemala
Honduras
El Salvador
Nicaragua
Costa Rica
Panama
St. Kitts & Nevis
Antigua and Barbuda
Dominica
St. Lucia
Barbados
Grenada
Trinidad & Tobago
Bermuda (U.K.)
Canary Islands (Spain)
Galapagos (Ecuador)
Hawaii Islands (U.S.)
French Polynesia
Easter (Chile)
Trindade (Brazil)
Ascension (U.K.)
St Helena (U.K.)
Falkland Islands (U.K.)
South Georgia (U.K.)
South Sandwich (U.K.)
Tristan da Cunha (U.K.)
Kiribati
Nauru
Tuvalu
Marshall Islands
Federated States of Micronesia
Palau
Solomon Islands
Vanuatu
Fiji
Tonga
New Caledonia (Fr.)
Wallis and Futuna (Fr.)
Samoa
Northern Mariana Islands (U.S.)
Guam (U.S.)
Coral Sea
Tasman Sea
Tasmania
New Zealand
North Island
South Island
Great Australian Bight
Christmas (Aus.)
Cocos Is. (Aus.)
Andaman & Nicobar Is. (India)
Sumatra
Java
Arctic Ocean
Atlantic Ocean
Pacific Ocean
Indian Ocean
Southern Ocean
Antarctica
Antarctic Circle
Arctic Circle
Tropic of Cancer
Tropic of Capricorn
Equator
Prime Meridian
Barents Sea
Kara Sea
Laptev Sea
East Siberian Sea
Wrangel
Novaya Zemlya
Franz Josef Land
October Revolution
Kotelny
Jan Mayen (Norway)
Svalbard (Norway)
Norwegian Sea
Baffin Bay
Beaufort Sea
Hudson Bay
Labrador Sea
Gulf of Alaska
Gulf of Mexico
Caribbean Sea
Mediterranean Sea
North Sea
Black Sea
Caspian Sea
Aral Sea
Red Sea
Arabian Sea
Bay of Bengal
Sea of Okhotsk
Sea of Japan (East Sea)
East China Sea
Philippine Sea
Bering Sea
Weddell Sea
1. Netherlands
2. Belgium
3. Luxembourg
4. Switzerland
5. Slovenia
6. Croatia
7. Bosnia and Herzegovina
8. Czechia
9. Slovakia
10. Austria
11. Hungary
12. Serbia
13. Moldova
14. North Macedonia
15. Albania
16. Cyprus
17. Lebanon
18. Guinea-Bissau
19. Guinea
20. Ghana
21. Togo
22. Benin
23. Cameroon
24. Equatorial Guinea
25. Rwanda
26. Cambodia
27. Panama
28. Malawi
29. Liechtenstein
30. Montenegro
31. Kosovo
32. Palestinian Territories
33. St. Vincent & the Grenadines
0
2,000
4,000 Miles
0
2,000
4,000
6,000
8,000 Kilometers
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AFRICA
INDIA
Fossil evidence of the Triassic land reptile Lystrosaurus.
SOUTH AMERICA
AUSTRALIA
ANTARTICA
Fossil remains of Cynognathus, a Triassic land reptile approximately 3m long.
Fossil remains of the freshwater reptile Mesosaurus.
Fossils of the fern Glossopteris, found in all of the southem continents, show that they were once joined.

- The continental drift hypothesis was developed in the early part of the 20th century, mostly by Alfred Wegener. Wegener said that continents move around on Earth’s surface and that they were once joined together as a single supercontinent. While Wegener was alive, scientists did not believe that the continents could move.Find a map of the continents and cut each one out. Better yet, use a map where the edges of the continents show the continental shelf. That’s the true size and shape of a continent and many can be pieced together like a puzzle. The easiest link is between the eastern Americas and western Africa and Europe, but the rest can fit together too.

- Alfred Wegener proposed that the continents were once united into a single supercontinent named Pangaea, meaning all earth in ancient Greek. He suggested that Pangaea broke up long ago and that the continents then moved to their current positions. He called his hypothesis continental drift.

- Continental drift is the hypothesis that the Earth's continents have moved over geologic time relative to each other, thus appearing to have "drifted" across the ocean bed. The idea of continental drift has been subsumed into the science of plate tectonics, which studies the movement of the continents as they ride on plates of the Earth's lithosphere.

- The speculation that continents might have 'drifted' was first put forward by Abraham Ortelius in 1596. A pioneer of the modern view of mobilism was the Austrian geologist Otto Ampferer.
  The concept was independently and more fully developed by Alfred Wegener in 1912, but the hypothesis was rejected by many for lack of any motive mechanism.
- Arthur Holmes later proposed mantle convection for that mechanism

# Tectonic Activity

Scientists did not accept Wegener’s theory of continental drift. One of the elements lacking in the theory was the mechanism for how it works—why did the continents drift and what patterns did they follow? Wegener suggested that perhaps the rotation of the Earth caused the continents to shift towards and apart from each other. (It doesn't.)

- Today, we know that the continents rest on massive slabs of rock called tectonic plates. The plates are always moving and interacting in a process called plate tectonics.
- The continents are still moving today. Some of the most dynamic sites of tectonic activity are seafloor spreading zones and giant rift valleys.

- In the process of seafloor spreading, molten rock rises from within the Earth and adds new seafloor (oceanic crust) to the edges of the old. Seafloor spreading is most dynamic along giant underwater mountain ranges known as mid-ocean ridges. As the seafloor grows wider, the continents on opposite sides of the ridge move away from each other.
- The North American and Eurasian tectonic plates, for example, are separated by the Mid-Atlantic Ridge. The two continents are moving away from each other at the rate of about 2.5 centimeters (1 inch) per year.

- Rift valleys are sites where a continental landmass is ripping itself apart. Africa, for example, will eventually split along the Great Rift Valley system. What is now a single continent will emerge as two—one on the African plate and the other on the smaller Somali plate. The new Somali continent will be mostly oceanic, with the Horn of Africa and Madagascar its largest landmasses